Multilevel factors influencing child overweight and obesity in the Pacific Islands

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Resumo (Abstract)

Summary This systematic review summarizes multilevel factors influencing overweight and obesity among children in the Pacific Islands from inception to April 2024. Factors assessed in studies were grouped by types of influence and levels of the social ecological model (SEM). 38 studies met inclusion criteria. Multilevel factors influencing overweight and obesity were caregiver factors (n = 22), geography (n = 20), sociocultural factors (n = 18), child diet (n = 10), child physical activity (n = 8), race and ethnicity (n = 8), infant factors (n = 7), other child health factors (n = 5), and environmental factors (n = 2). Majority of studies focused on factors at the individual level of the SEM (n = 33), followed by the interpersonal level (n = 21), and the community level (n = 20). No studies were found that address factors at the policy level of the SEM. There were some consistent findings for modern diet, screen time, maternal BMI, and type of school the child attended (public vs. private) being positively associated with child overweight and obesity. Funding 10.13039/100000199 USDA 2021-68012-35899, 10.13039/100000002 NIH 1P20GM139753-01A1, 10.13039/100000002 NIH T32DK137523, 10.13039/100000002 NIH K99HL166781.

Introduction

In recent decades, Pacific island countries and territories (PICTs) have seen major shifts in ways of life, and globalization has influenced the lifestyle and health status of island communities. 1 , 2 Pacific populations have seen an increase in chronic disease morbidity, particularly with the rise of overweight or obesity (OWOB) across generations. 3 , 4 , 5 Childhood obesity has many adverse health consequences throughout the life course. 6 , 7 , 8 , 9 Obesity in childhood tracks to adulthood and is strongly associated with major chronic diseases. 6 , 10 Not only can it increase the risk for developing harmful diseases during childhood and adolescence, children and adolescents who have OWOB are more likely to develop chronic diseases as adults, such as type 2 diabetes, cardiovascular disease, asthma, and various cancers. 4 , 6 , 7 , 8 Additionally, childhood and adolescent OWOB has been linked to adverse psychosocial outcomes, such as attention deficit hyperactivity disorder (ADHD), sleep abnormalities, and low self-esteem. 9 , 11

Recent studies show that while childhood obesity has been increasing worldwide over the past several decades, with current prevalence in PICTs 4 , 12 currently estimated at 24–30% of children affected. 4 A global study, which assessed body mass index (BMI) of 130 million children and adolescents from 1975 to 2016, found childhood and adolescent BMI increased 0.32 kg/m 2 per decade for girls and 0.4 kg/m 2 per decade for boys. 4 Although there has been an overall increase in BMI worldwide, developed countries such as the United States (U.S.), Australia, and countries in Western Europe, have reached a plateau in BMI, and countries in Eastern Europe had no increase in BMI over this time period. 4 In contrast, Pacific island regions of Polynesia and Micronesia have had the greatest continuous increase in childhood and adolescent BMI over this period. A separate global study, assessing the dual burden of underweight and obesity trends from 1990 to 2022, found a substantial rise in childhood obesity in Pacific island countries. By 2022, obesity rates far exceeded

those of underweight in this region. 12 Although currently undernutrition and overnutrition co-exist, preadolescent child obesity is a larger proportion of the burden compared to undernutrition in most islands. 12

To design effective prevention and intervention strategies, a multilevel understanding of childhood obesity determinants is essential. These factors can influence children at multiple socio-ecological levels. These include factors at the individual level such as diet preferences, physical activity level, screen time; factors at the interpersonal or familial level such as socio-cultural values, acculturation, family expectations, parental/caregiver education or literacy, socioeconomic status, knowledge of food and disease, risk taking behavior; and factors at the community or environmental level such as location of residence, limited availability and accessibility to nutritious foods or health promoting environments, and lack of health and wellness policies. 11 , 13 , 14 Few systematic reviews have focused on factors influencing child overweight and obesity in the Pacific. 15 , 16 , 17 , 18 , 19 , 20 , 21 , 22 , 23 , 24 This paper reviews the multilevel factors influencing obesity in preadolescent children in 22 PICTs.

Methods

The systematic review followed methodology and reporting guidelines from the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 25 , 26 and Joanna Briggs Institute Manual for Evidence Synthesis. 27

Eligibility criteria

Eligible studies included children at birth and up to age 12 years, measured anthropometric OWOB outcomes, factors for OWOB, and PICTs within the Western Pacific region as defined by the World Health Organization (WHO) and classified by the Pacific Community (SPC) Pacific Standard Industrial Classification. 28 This included 22 PICTs within Polynesia (American Samoa, Cook Islands, French Polynesia, Niue, Pitcairn Islands, Samoa, Tokelau, Tonga, Tuvalu, Wallis and Futuna), Micronesia (Commonwealth of the Northern Mariana Islands (CNMI), Federated States of Micronesia (FSM, inclusive of Chuuk, Yap, Kosrae, Pohnpei), Guam, Kiribati, Marshall Islands, Nauru, Palau), and Melanesia (Fiji, New Caledonia, Papua New Guinea, Solomon Islands, Vanuatu). 28 Hawai'i, Aotearoa (New Zealand) and Rapa Nui (Easter Island) were part not of the SPC classification and thus excluded from this review. Letters to the editors, commentaries, case studies, protocols, reviews, and publications written in a language other than English or French were excluded. Intervention studies, including randomized controlled trials, and studies reporting only OWOB by age and sex (the latter due to the focus on prevalence in paper 1 of this series) were excluded unless there were other reported determinants for OWOB reported at baseline. Qualitative studies were excluded due to having no measures of primary outcomes of interest. There were no restrictions on publication date.

The primary outcomes of interest were OWOB assessed by BMI category, standardized BMI score (BMI z-score), macrosomia status, large for gestational age, or other anthropometric measures with an OWOB cut-point identified (e.g., body fat percentage, waist circumference, hip-to-waist ratio, skin folds, macrosomia, or large for gestational age).

Search strategy

The literature search strategy was developed using the following concepts: (1) OWOB, (2) children, (3) anthropometry measures, (4) epidemiological or risk factors for OWOB, and (5) PICT populations. The initial search strategy was developed by author AO. The search strategy is listed in Table 1 . This

review was registered with PROSPERO (2025 CRD420251090452). A systematic review protocol was developed to guide this review and can be found online via PROSPERO. Table 1 Search terms & details. Study items Search terms & details Condition or domain English: Obes, a Overweight French: obés, a surpoids, a adipos, a indice de masse corporelle, IMC, poids a Participants English: Infant, a Child, a Pediatr, a Paediatr a (with a median age between birth and 12 years.) French: enfant, a pédiatri a Exposures English: Determinants, factors, epidemiological factors, risk factors, diet, nutrition, physical activity, digital environment, screen time, economic, social, commercial, sleep, education, income, literacy, environment, hous, a store, food, park, play French: déterminant, a facteur Population locations English: American Samoa, Cook Islands, Federated States of Micronesia, Chuuk, Yap, Kosrae, Pohnpei, Ponape, Fiji, French Polynesia, Guam, Kiribati, Nauru, New Caledonia, Niue, Commonwealth of the Northern Mariana Islands, Marshall Islands, Palau, Papua New Guinea, Pitcairn Islands, Samoa, Solomon Islands, Tokelau, Tonga, Tuvalu, Vanuatu, Wallis and Futuna French: iles du Pacifique a Indicates truncated search term in which multiple variations of the word was searched.

Searches for studies in English were conducted in April 2024 using the following databases: PsycINFO, CINAHL, EMBASE, EBSCO/AGRICOLA, PROQUEST (includes ERIC), Web of Science, MEDLINE/PUBMED- MeSH, Scopus, SPORTDiscus. Backward and forward citation chasing on included articles from the initial English search was executed in September 2024 using citationchaser, 29 SpiderCite 30 and Research Rabbit 31 software to identify additional relevant studies. An additional search was conducted for studies in French from inception up to and including November 2024 using Google Scholar with Publish or Perish software as French literature could not be searched using the aforementioned databases. Hand searching was not conducted. Gray literature was not searched to limit the review to published academic literature with robust study methods.

Data management

PRISMA Checklist and Flow Diagram were used to guide the screening process. 25 , 26 All search results from databases were imported into Covidence 32 to deduplicate and screen each article. Other relevant studies identified in citation chaining were added to the Covidence project. After eliminating duplicates, a three-stage process was conducted to screen articles. First, titles and abstract of identified articles were screened based on the eligibility criteria. Articles addressing non-relevant participants or populations, editorials or reviews, and articles reporting on RCTs were excluded. Then, the remaining full text of identified articles were reviewed, and inclusion/exclusion criteria were applied again. All articles identified from databases were reviewed by two reviewers with a third reviewer as the final consensus for any discrepancies determined. Articles found from citation chaining were reviewed by a single reviewer. Finally, data were extracted by two reviewers with consensus by a third reviewer, as needed. The authors used a customized data-extraction sheet to collect data and relevant information that were consistent with the goal of the systematic review, including study design, year(s) of study or data used, population, location and setting, age group, race and ethnicity, anthropometric measurements, and outcomes.

For every eligible study, risk of bias was assessed with JBI critical appraisal tools in Qualtrics. 27 , 33 All eligible studies were evaluated on specific criteria according to their study design type (cross-sectional or cohort). Cross-sectional studies were evaluated based on 8 criteria: 1) study subjects & setting described in detail, 2) inclusion criteria clearly defined, 3) objective standard criteria were used to measure condition, 4) exposures measures were valid and reliable, 5) confounding factors identified, 6) strategies to deal with confounding were stated, 7) outcome measures were valid

and reliable, and 8) appropriate statistical analysis were used. Cohort studies were evaluated based on 12 criteria, five of which were the same as cross-secitonal criterias 4–8 and included seven additional criteria: 9) two groups were similar and recruited from the same population, 10) exposures were measures similarly in both exposed and unexposed groups, 11) groups/participants were free of the outcome at the start of the study, 12) outcome measures were valid and reliable, 13) follow up time was reported and sufficient to be long enough for outcomes to occur, 14) follow up was completed and if not, reasons were explained, and 15) strategies to address incomeplete follow up were utilized. Studies that met at least 75% of criteria were considered low bias. Studies that met less than 75% of criteria were considered high bias.

Presentation of findings

Characteristics of eligible studies were mapped by location and described by sample size, age range, year of data, anthropometric measurements, key factors associated with childhood OWOB in the Pacific, and risk of bias. Eligibility required description of anthropometry that included a determination of obesity. These included child obesity organized into categories using WHO child growth standards or references, 34 , 35 Centers for Disease Control and Prevention (CDC) growth charts, 36 International Obesity Task Force (IOTF) criteria, 37 and other country-specific references. These definitions include: overweight and obesity are defined as BMI for age and sex >85th percentile (overweight) or > 95th percentile (obese) according to CDC for children, and >1 standard deviations (overweight) or >2 standard deviations (obese) above BMI z score for age and sex according to WHO for children over 5 y. Macrosomia was defined as >4000 g at birth.

Outcome measures were grouped by primary outcomes (anthropometric measures related to OWOB, including BMI, BMI z-score, Weight for age, and weight for length) and by determinant factor type. Key factors identified within studies that were assessed by OWOB outcomes were organized into broader factor categories. Factor categories were grouped within the social ecological model (SEM) to better understand their influence on child OWOB at 3 SEM levels: 1) individual, which included factor categories related to demographics (e.g., age and sex) and behavioral factors (e.g., dietary intake and physical activity); 2) interpersonal, which included factor categories related to caregivers (e.g., primary caregiver health status) and sociocultural environment (e.g., household socioeconomic resources); and 3) community which included factor categories related to the child's environment (e.g., school, neighborhood)and geography (e.g., island residence, urbanicity). 38 This framework allows for a structured interpretation of the factors contributing to childhood OWOB in the Pacific Islands.

Results

A total of 4985 citations were retrieved from the search, from which 1452 duplicates were removed. After screening titles and abstracts, 396 full texts were examined and 35 articles met the eligible criteria and were included. Citation chaining of these included articles identified 2062 more articles and an additional search of French articles identified 196 articles. After screening titles and abstracts found from these additional search methods, 141 full text articles were reviewed, which identified 3 additional articles that met the eligible criteria. Thus, a total of 38 articles or individual studies were included in this review (Fig. 1). Fig. 1 PRISMA flow chart .

Thirty of the 38 eligible studies were cross-sectional and seven were cohort studies (Table 2). Most studies were relatively recent with data from 2000 to 2024. Two studies were older having been conducted in the 1970s. Studies included children across all ages from 0 to 12 years old, with all

cohort studies including only infants, toddlers aged 0–2 years old, or young children aged 2–6 years old (Table 2). These studies provide the strongest evidence of causal relationship between factors and obesity. Maternal age, gestational diabetes, high BMI, lower education and higher wealth indices predicted infant macrosomia and OWOB (Table 2). Table 2 Characteristics of studies that assessed factors influencing childhood overweight or obesity (OWOB) in the Pacific Islands. Study Location Sample size Age range Year of data Anthropometry measure (Reference, if stated) Key factors Cross-Sectional Alur et al. (2002) 39 Guam 332 0 years (infants at birth) 2000–2002 Birthweight, Large for Gestational Age Race/Ethnicity, Maternal Diabetes Bindon et al. (1986) 40 American Samoa, Samoa 786 5.5–11.5 years 1975–1982 BMI (HANES I values from Cronk and Roche, 1982, 352, 353) Age, Sex, Residence Blankenship et al. (2020) 41 Marshall Islands 729 0–59 months 2017 Weight-for-Height, Weight-for-Age (WHO) Age, Maternal BMI Choy et al. (2017) 42 Samoa 305 24–59 months 2015 BMI z score (WHO) Age, Sex, Diet (breakfast, fruits, vegetables, sweetened fruit juice, frozen desserts, total energy, total protein, total carbohydrates, total fat, monounsaturated fatty acids, polyunsaturated fatty acids, saturated fat, fiber, sugar, cholesterol, sodium, vitamin A, retinol, beta carotene, vitamin E, vitamin C, iron, calcium, potassium), Ever Breastfed, Physical Activity, Screen Time, Sleep Hours, Material Lifestyle score, Household Income, Birth Order, Community Spirit, Census Region, Birthweight, Illness in the past 3 weeks, Deworming Medication Use, Maternal Age, Maternal Education, Maternal Marital Status, Maternal OWOB, Maternal Anemia Choy et al. (2018) 43 Samoa 305 24–59 months 2015 BMI (WHO) Diet (modern vs. neo-traditional) Choy et al. (2023) 44 Samoa 437 5–11 years 2019–2020 BMI, Weight-for-Age, Height-for-Age (WHO) Age, Sex, Residence Urbanicity, Census Region Daigre et al. (2012) 45 French Polynesia 145 5–14 years 2007–2008 BMI z score; BMI percentiles; Waist Circumference (WHO, IOTF & French Reference) Age, Sex, Residence Dancause et al. (2012) 46 Vanuatu 577 0–17 years old 2007 BMI z score, Weight-for-Age z score, Weight-for-Length z score, Waist Circumference, Waist-to-Height Ratio (WHO) Age, Sex, Residence Frayon et al. (2020) 47 New Caledonia 256 10–16 years 2015–2016 BMI Status (IOTF) Age, Sex, Race/Ethnicity, Residence Urbanicity, Income Fuka et al. (2020) 48 Fiji 255 0 years (infants at birth) 2013–2014 Birthweight Maternal Age, Maternal Ethnicity, Maternal Education, Maternal Marital Status, Gravidity, Family History of Diabetes, Previous History of GDM, Previous baby >4 kg, Previous stillbirth, Previous neonatal death, Maternal BMI, Booking age of gestation, Gestational age at delivery, Induced Labor, Caesarean section, Preeclampsia Fukuyama et al. (2005) 49 Tonga 177 5–19 years 2002–2003 BMI, Triceps Skinfold (ITOF & CDC) Age, Sex, Residence Gibson et al. (2011) 50 Tonga 316 0–18 years 2005–2006 BMI, Weight-for-Age (WHO) Age, Migration status Hawley et al. (2014) 51 American Samoa 1053 0–15 months 2001–2008 Weight-for-Length, Macrosomia Age, Sex, Infant Feeding Haycraft et al. (2023) 52 Fiji, Kiribati, Tuvalu 2662 0–24 months 2015–2021 Weight-for-Height z score (WHO) Residence (Slum) Heinsberg et al. (2020) 53 Samoa 399 3–7 years 2017 BMI, Abdominal Circumference Age, Sex, Bisphenol-A (BPA) exposure, Diet (More modern, Less modern), Physical Activity (More active than peers, About the same as peers, Less active than peers), Household Assets score, Annual Household Income Ho et al. (2021) 54 Papua New Guinea, Vanuatu 600 3–5 years 2012–2014 Weight-for-Height z score (WHO & Multicenter Growth Reference Study) Residence Hughes et al. (2004) 55 American Samoa, Cook Islands, FSM(Chuuk, Kosrae, Pohnpei, Yap), Fiji, French Polynesia, Kiribati, Nauru, Niue, Marshall Islands, Samoa, Solomon Islands, Tonga, Tuvalu, Vanuatu 3826 5–12 years 2001–2002 BMI z score, Weight-for-Age z score, Weight-for-Height z score (WHO) Sex, Country Residence, Helminthiasis parasite status, Hemoglobin levels Kemmer et al. (2009) 56 American Samoa 208 5–10 years 2003 BMI z score, Weight-for-Age z score (CDC) Age, Sex, Anemia Leon Guerrero et al. (2020) 57 Guam 865 2–8 years 2013 BMI, BMI z score, BMI status, Waist Circumference, Abdominal Obesity (CDC & International Diabetes Foundation) Age, Sex,

Race/Ethnicity, Diet (water, milk, sugar sweetened beverage, fruit, vegetable), Infant Feeding, Age weaned from breastmilk, Screen Time, Sleep, (hours, disturbance), Income, Food Assistance, Food Security, Caregiver Education, Household Structure (number children in household), Parent Marital Status, Birth Place, Acculturation Status Mahumud et al. (2021) 58 Cook Islands, Fiji, Kiribati, Nauru, Niue, Samoa, Solomon Islands, Tokelau, Tonga, Tuvalu, Vanuatu, Wallis & Fatuna 14,469 11–17 years 2003–2015 BMI status (WHO) Sex, Diet, (Fruit, Vegetable, Carbonated soft drink, Fast food), Physical Activity minutes, Sedentary Activities, Island Residence Novotny et al. (2016) 59 American Samoa, CNMI, FSM (Chuuk, Kosrae, Pohnpei, Yap), Guam, Marshall Islands, Palau 5558 2–8 years 2011–2013 BMI (CDC) Residence Novotny et al. (2022) 60 American Samoa, CNMI, FSM, (Chuuk, Yap, Kosrae, Pohnpei),Guam, Marshall Islands, Palau 3520 2–8 years 2013 BMI status (CDC) Diet (grains, fruits, vegetables, milk, meats, total protein, total carbohydrates, total starch, total fat,omega-3, omega-6, monounsaturated fatty acids, polyunsaturated fatty acids, saturated fat), World Bank Income Status, Household Income, Island Residence Passmore et al. (2019) 61 Marshall Islands 3271 4–16 years old 2017–2018 BMI Status (CDC) Age, Sex, School Type Paulino et al. (2017) 62 FSM (Chuuk, Kosrae, Pohnpei, Yap), Marshall Islands, Palau 1201 2–8 years 2013–2015 BMI Status (CDC) Household with caregiver chewing betel nut Pham et al. (2021) 63 Papua New Guinea 3132 0–5 years 2018 Weight-for-Age Age, Sex, Household Wealth, Household Food Shortage, Caregiver Education, Residence, Residence, Urbanicity Thompson et al. (2020) 64 Samoa 83 3–7 years 2015–2017 BMI Status, Weight-for-Age, Percent Body Fat, Percent Lean Mass, Trunk-to-peripheral fat ratio, DXA, Lunar Trunk-to-Peripheral Fat Ratio (WHO & IOTF) Age, Sex, Diet (protein, carbohydrates, fat, total energy), Physical Activity (accelerometer), Material Lifestyle score Tsitas et al. (2015) 65 , a Samoa 0 years (infants at birth) 2007–2007 Birthweight Sex, Race/Ethnicity, Delivery Mode, Gestational Diabetes Mellitus; 5′ APGAR, NICU Transfer Tubert-Jeannin et al. (2018) 66 New Caledonia 1155 6, 9, & 12 years 2011–2012 BMI status, Waist-to-Height Ratio (WHO & IOTF) Age, Sex, Race/Ethnicity, Diet (sweet drinks), Geographic area, Type of School, Toothbrushing frequency, Smoking habits, Dental Attendance, Prevention at School, Dental Anxiety, School Lunch Smith et al. (2007) 67 Tonga 443 11–16 years 2000 BMI status (Polynesian cut off points) Sex, Diet (Fruits, Soft drinks, Sweets/chocolates, Fresh vegetables, Taro, Tinned fish, Tinned mutton/corned beef), Exercise frequency/duration, Screen Time, Residence, Caregiver employment Warmath et al. (2024) 68 Samoa 445 3–8 years 2017–2018 BMI z score (WHO) Physical Activity (Highly Active) Cohort Choy et al. (2022) 69 Samoa 197 2–6 years 2015–2017 BMI z score (WHO) Age, Sex, Diet (total energy intake, iron, calcium, vitamin C), Ever Breastfed, Physical Activity, Screen time, Sleep hours, Household Asset score, Household Income, Census region, Child Anemia, Birthweight, Maternal Age, Maternal Education, Maternal Marital Status, Maternal BMI, Maternal Anemia Dela Cruz et al. (2018) 70 CNMI 8427 0 years (infants at birth) 2007–2014 Birthweight Maternal Age, Maternal Race/Ethnicity, Maternal Number of Antenatal Care Visits Hawley et al. (2015) 71 American Samoa 1036 0–12 months 2001–2008 Infant size at 12 months, Weight-for-Length, Large for Gestational Age (CDC) Age, Gestational weight gain, Maternal age, Parity, Pregnancy weight at 13 weeks, Birthweight Osuagwu et al. (2020) 72 Fiji 255 0 years (infants at birth) 2013–2014 Birthweight Race/Ethnicity, Maternal Age, Maternal Education, Marital Status, Gravidity, Maternal Family history of diabetes, Maternal history of GDM, Previous macrosomia baby, Previous stillbirth, Previous neonatal death, Maternal BMI, Gestational Age at Visit, Gestational Age at GDM Diagnosis, Gestational Age at Delivery Oyama et al. (2023) 73 Samoa 160 0–24.5 months Not Specified Weight-for-Length z score, Weight-for-Height z score (WHO) Age, Household Assets Pelletier et al. (1986) 74 American Samoa 1186 0–12 months 1974–1976 Weight-for-Age, Weight Gain Sex, Infant Feeding Sugiyama et al. (2017) 75 Palau 1730 0 years (infants at birth) 2007–2014 Birthweight Maternal Gestational Diabetes Mellitus Therrien et al. (2020)

76 Vanuatu 54 <1 year 2016–2017 BMI z score (WHO) Age, Sex, Birthweight, Distress Postpartum, Maternal Dietary Diversity, Pregnancy Distress a Article labeled study design as case control though description of design was cross-sectional and thus assessed as a cross-sectional study.

Cross-sectional studies included all ages with some focusing only on infants or narrow age groups while others included wide age ranges from 0 to 12 years (Table 3). At least 1 study was conducted in all but 1 eligible PICT (Fig. 2 , Table 1). No studies that assessed factors of childhood OWOB were found to be conducted in Pitcairn Islands. About half the studies were conducted in or included Samoa (n = 12) and American Samoa (n = 8); 6 studies included the Marshall Islands; 5 studies included Fiji, Tonga and/or Vanuatu; 4 studies included FSM, Palau, and/or Guam; 3 studies included CNMI, Kiribati, and/or Tuvalu; 2 studies included Nauru, Niue, New Caledonia, PNG, Solomon Islands, Cook Islands, and/or French Polynesia; and 1 study included Wallis Futuna and Tokelau. Table 3 Summary of findings on factors with higher (↑), lower (↓), or no difference (–) on childhood overweight or obesity (OWOB), body mass index (BMI), or macrosomia. Type of findings Key factors or indicator Mixed findings Consistent findings Single study findings Study design Cross-Sectional Cohort Cross-Sectional Cross-Sectional Cohort Factor category Demographics Age Any Age: ↑[41 b , 44 a , 45 , 46 , 49 – 51 b , 53 a , 56 , 61 b , 64 , 66] ↓[42 , 47 b] –[63 b] Any Age: ↓[76] –[69] By Age Groups: ↑ 0–1 yr: [41 b , 51 b] 2–7 yrs: [44 a , 53 a , 64] 4–12 yrs: [61 b] 5–12 yrs: [45 , 49 , 56 , 66] 0–12 yrs: [46 , 50] ↓ 2–5 yrs: [42] 9–12 yrs: [47 b] – 0–5 yrs: [63 b] By Age Groups: ↓ 0–1 yrs: [76] – 2–6 yrs: [69] Sex Boy vs. Girl: ↑[42 , 44 , 51 b , 53 , 55 , 58 , 61 b , 64 , 65 b , 67] ↓[49 a , 56 a , 66] –[45 , 46 , 47 b , 63 b] Boy vs. Girl: –[69 , 74 b , 76] Race and Ethnicity c Pacific Islander vs. Other: ↑[39 b , 47 b , 57 , 65 b , 66 a , d] ↓[48] Pacific Islander vs. Other: ↑[70], ↓[72] Child behavior: Diet Food Groups Fruits/Vegetables: ↑[42 , 57], ↓[58 a , 67], –[60] SSB: ↑[57 , 58 , 67], ↓[66] Grains: ↑[60 a] Milk: ↓[60 a] Macronutrients Carbohydrates: ↓[42 , 64 a], –[60] Total Fat: –[60 , 64], ↑[42] Energy: –[42 , 64] Protein: –[42 , 60 , 64] Energy: ↑[69] Traditional vs. Modern Modern: ↑[43] Neo-Traditional: ↓[43] Less Modern vs. More Modern: ↓[53] BPA exposure ↑[53] Child behavior: Physical activity Activity Level ↑[68 a], ↓[42 , 53 , 64 a] ↓[69] Screen Time ↑[42 , 67] ↑[69] Sleep Hours ↑[42] –[69] Child behavior: Infant factors Birthweight ↑[71 a , 76] ↑[42] –[69] Feeding Type Breastfeeding: ↑[57 , 74] ↓[42 , 51 b] Formula: ↑[51 b] Mixed: ↓[51 b] Breastfeeding: ↓[69] Child behavior: Health conditions Anemia ↑[42 , 56], ↓[55 a] ↑[69] Recent Illness ↑[42] Helminthiasis ↓[55 a] Caregiver factors Maternal Age ↑[70], –[71] ↑[42] ↑[69] Marital Status –[42] ↑[69] Maternal BMI ↑[69], ↓[72] ↑[41 b , 42 , 48] GDM ↑[48], ↓[65] ↑[75 a] Preeclampsia ↑[48] Maternal Diabetes ↑[39 b] Maternal Diet ↓[76] Other Caregiver Behaviors Chews Betel Nut: ↑[62] Sociocultural Economic Status (Income, Wealth, Assets, Education) Household Assets: ↑[53 a , 64] Material Lifestyle Score: ↑[42] Income: ↑[42 , 63 b], ↓[53] Socioeconomic status: ↓[47 b] Caregiver Education: ↑[42 , 57] Household Assets: ↑[69 , 73] Maternal Education: ↓[72] Income: ↓[69] Caregiver Education: ↑[69] Food Security Food Assistance: ↑[57] Food Shortage: ↓[63 b] Acculturation Integrated vs. Traditional: ↑[57] School Type Public vs. Private: ↓[61 b , 66] Geography e Urbanicity Urban vs. Rural: ↑[42 , 44], ↓[47 b , 63] Urban vs. Rural: ↑[69] Slum Residence ↓[52 b] Migration Status ↑[50] Birthplace Guam vs. Not Guam: ↑[57] Environment Community Spirit –[42] Prevention at School –[66] Numbers listed refer to study reference number. a Statistically significant finding. b Descriptive finding with no statistical test applied. c Pacific Islander includes any race or ethnicity group considered Pacific Islanders (e.g. Chamorro, Fijian, Micronesian, Polynesian, etc.). d Significant results only for Polynesian subgroup. e Island/Village Residence findings not included in table due to variations in comparison groups among studies. Fig. 2 Number of eligible studies that examined factors of childhood overweight or obesity in each Pacific island country or territory (Map from https://www.freeworldmaps.net) . 77

Of the 12 studies from Samoa, 7 were part of the Ola Tuputupu'ae “Growing Up” program. Four studies that included American, Guam, CNMI, FSM, Palau, and/or Marshall Islands were part of the Children's Healthy Living (CHL) program. One study from Tonga was part of the Health Behaviour and Lifestyle of Pacific Youth (HBLPY) program. Other studies were part of the Majuro Youth Lifetime Health program (Marshall Islands), Foafoaga O le Ola (Samoa), or Healthy Mothers, Healthy Communities (Vanuatu). A study that utilized the Global School-based Student Health Survey (GSHS) included 12 PICTs and another that utilized the East Asia Pacific Early Child Development Scales (EAP-ECDS) included 2 PICTs. Other studies utilized the PODIUM survey (French Polynesia) and Comprehensive Health and Epidemiological Surveillance System (CHESS) (PNG).

The reported factors influencing childhood OWOB in PICTs were diverse. Six factor categories emerged from the studies (Fig. 3): 1) Demographic factors such as age (n = 21), sex (n = 22), and race and ethnicity (n = 8); 2) Behavioral factors such as child feeding and diet (n = 10), child physical activity (n = 8), infant factors (n = 7), other child health conditions (n = 5); 3) Socio-cultural factors such as economic status (n = 11); household structure (n = 2), food assistance (n = 2), acculturation (n = 1), household betel nut chewing (n = 1); 4) Maternal factors such as caregiver demographics (n = 9), maternal BMI (n = 5), maternal diet (n = 1), maternal gestational diabetes (GDM) (n = 2), preeclampsia (n = 1), and clinical outcomes (n = 3); 6) Geographical factors such as residence (n = 14), birthplace (n = 1), urbanicity (n = 3), migration status (n = 1), and slum (n = 1); and 6) Environment factors such as BPA exposure (n = 1), obesity prevention exposure at school (n = 1), and community spirit (n = 1). Fig. 3 Factors influencing childhood obesity in the Pacific Islands organized by social ecological model (SEM) level (Individual, Interpersonal, Community) and factor category (Behavioral, Demographics, Sociocultural, Caregiver, Geography, and Environment). Highly studied factors shown by frequency (n) are highlighted in bold. Yellow highlighted factors are positively associated with child overweight and obesity .

Majority of studies focused on factors at the individual level of the SEM (n = 33), followed by the interpersonal level (n = 21), the community level (n = 20) (Fig. 3). No studies addressed factors at the policy level of the SEM.

Mixed findings were identified for most of the key factors, while there were some consistent positive associations for indicators of modern diet, screen time, maternal BMI, and type of school the child attended (public vs. private) (Table 3 , Supplemental Table S1). Children who consumed a more modern diet, 43 , 53 had more screen time, 42 , 67 , 69 had mothers with higher BMI, 41 , 42 , 48 , 69 and attended public school 61 , 66 were more likely to have OWOB. In more studies, higher BMI or OWOB prevalence was found among children of increasing age, 41 , 44 , 45 , 46 , 49 , 50 , 51 , 53 , 56 , 61 , 64 , 66 boys, 42 , 44 , 51 , 53 , 55 , 58 , 61 , 64 , 65 , 67 Pacific Islander race and ethnicities, 39 , 47 , 57 , 65 , 66 those from higher socioeconomic status (SES), 42 , 53 , 57 , 63 , 64 , 69 , 73 and who had less physical activity 42 , 53 , 64 , 67 , 69 ; however, these findings were still mixed.

Most factors examined were assessed in young children ages 2–7 years old (Table 4), which included age, sex, economic status, island or village residence, diet, physical activity, infant factors, or caregiver demographics. Influences of age, sex, and residence were assessed in the majority of studies at all ages, 0–12 years. Several studies assessed caregiver demographics and clinical factors in infants at birth. Table 4 Heat map presenting number of studies by age of participants and key factors within each social ecological model (SEM) level influencing childhood overweight or obesity in the Pacific (total n = 38 studies). Color represents SEM level and shade of color represents number of eligible studies assessing participants at the age in years (darker shades represent more studies including participants at that age).

The cross-sectional studies generally had low bias, yet potential confounders were not always identified and accounted for (Table 5). In cohort studies, loss to follow up was not always explained. These factors limit conclusiveness of findings. Table 5 Quality assessment of studies using the Joanna Briggs Institute (JBI) critical appraisal tools. a Author, Year JBI appraisal criteria Overall appraisal b Exposure measures were valid & reliable with standard criteria used for all groups b Outcome measures were valid & reliable b Confounding factors identified b Confounding factors accounted for b Appropriate statistical analysis used b Study subjects, sample inclusion criteria, & setting well described c Participants free of the outcome at the start of study d Sufficient follow up time provided d Loss to follow up addressed d Groups were comparable or similar d Cross-Sectional Alur et al. (2002) 39 Low Bias ✓ ✓ X X ✓ ✓ n/a n/a n/a n/a Bindon et al. (1986) 40 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Blankenship et al. (2020) 41 Low Bias ✓ ✓ X X ✓ ✓ n/a n/a n/a n/a Choy et al. (2017) 42 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Choy et al. (2018) 43 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Choy et al. (2023) 44 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Daigre et al. (2012) 45 Low Bias ✓ ✓ X X ✓ ✓ n/a n/a n/a n/a Dancause et al. (2012) 46 Low Bias ✓ ✓ ✓ X ✓ ✓ n/a n/a n/a n/a Frayon et al. (2020) 47 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Fuka et al. (2020) 48 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Fukuyama et al. (2005) 49 Low Bias ✓ ✓ X ✓ ✓ ✓ n/a n/a n/a n/a Gibson et al. (2011) 50 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Hawley et al. (2014) 51 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Haycraft et al. (2023) 52 Low Bias ✓ ✓ X X ✓ ✓ n/a n/a n/a n/a Heinsberg et al. (2020) 53 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Ho et al. (2021) 54 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Hughes et al. (2004) 55 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Kemmer et al. (2009) 56 Low Bias ✓ ✓ X X ✓ ✓ n/a n/a n/a n/a Leon Guerrero et al. (2020) 57 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Mahumud et al. (2021) 58 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Novotny et al. (2016) 59 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Novotny et al. (2022) 60 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Passmore et al. (2019) 61 Low Bias ✓ ✓ ✓ X ✓ ✓ n/a n/a n/a n/a Paulino et al. (2017) 62 Low Bias ✓ ✓ X X ✓ ✓ n/a n/a n/a n/a Pham et al. (2021) 63 Low Bias ✓ ✓ X X ✓ ✓ n/a n/a n/a n/a Thompson et al. (2020) 64 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Tsitas et al. (2015) 65 , e Low Bias ✓ X ✓ X ✓ ✓ n/a n/a n/a n/a Tubert-Jeannin et al. (2018) 66 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Smith et al. (2007) 67 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Warmath et al. (2024) 68 Low Bias ✓ ✓ ✓ ✓ ✓ ✓ n/a n/a n/a n/a Cohort Choy et al. (2022) 69 Low Bias ✓ ✓ ✓ ✓ ✓ n/a X ✓ ✓ ✓ Dela Cruz et al. (2018) 70 Low Bias ✓ ✓ ✓ ✓ ✓ n/a ✓ ✓ X ✓ Hawley et al. (2015) 71 Low Bias ✓ ✓ ✓ ✓ ✓ n/a ✓ ✓ ✓ ✓ Osuagwu et al. (2020) 72 Low Bias ✓ ✓ ✓ ✓ ✓ n/a ✓ ✓ X ✓ Oyama et al. (2023) 73 Low Bias ✓ ✓ ✓ ✓ ✓ n/a ✓ ✓ X ✓ Pelletier et al. (1986) 74 Low Bias ✓ X ✓ ✓ ✓ n/a ✓ ✓ X ✓ Sugiyama et al. (2017) 75 Low Bias ✓ ✓ X X ✓ n/a ✓ ✓ X ✓ Therrien et al. (2020) 76 Low Bias ✓ ✓ ✓ ✓ ✓ n/a ✓ ✓ ✓ ✓ a JBI survey tools for each study design differed. The table below consolidates criteria for cross-sectional and cohort studies. b Criteria assessed for both cross-sectional and cohort studies. c Criteria assessed for only cross-sectional studies. d Criteria assessed for only cohort studies. e Article labeled study design as case control though description of design was cross-sectional and thus assessed as a cross-sectional study.

Discussion

Research on childhood OWOB in PICTs has grown in recent decades, and this review provides a comprehensive synthesis of multilevel factors contributing to OWOB in this region. Prior to 2000, studies in the Pacific were focused on undernutrition, which was more prevalent than overnutrition at that time. While early research focused largely on undernutrition, the shift toward overnutrition necessitates a timely understanding of contributing factors. Studies on this topic began to emerge in the new millennium alongside the establishment of research programs aimed at understanding childhood OWOB. These programs utilize comprehensive survey tools and build ongoing community

partnerships. Despite this, the body of evidence of childhood OWOB factors in the Pacific is still limited in comparison to research available in other regions. Within the PICT, there is also a disparity where some countries have more research available than others. This disparity may reflect varying prevalence of OWOB, differences in research capacity and health infrastructure, or funding available across PICTs. In particular, French speaking territories in the Pacific are under-represented, which may reflect language barriers in publication or limited access of data. 78 However, as the nutrition transition spreads across all Pacific islands, understanding key factors and their patterns of emergence in the PICT, which represent low to high income countries would be beneficial for preventative efforts.

The majority of studies were cross-sectional and more longitudinal studies would provide stronger evidence to infer causality of factors influencing childhood obesity. The evidence from the studies included in this review are limited in that many of the studies were descriptive and confounders were not always accounted for. Additional studies exist in the wider Pacific region and its periphery, including Hawai'i, Aotearoa (New Zealand), Rapa Nui (Easter Island), Australia, the United States of America, and future investigations can extend the scope of our work to include these settings where Pacific people live and children are also affected by OWOB. There is a critical need for longitudinal research in the region to better identify multilevel predictors of childhood OWOB.

Several studies showed that maternal size is positively associated with infant birth weight and size in the first 2 years of life. 41 , 42 , 48 , 69 Clearly maternal health is a target for intervention prior to and during pregnancy, supporting a life course approach to obesity prevention. More modern diet, 43 , 53 screen time, 42 , 67 , 69 and public school attendance 61 , 66 were also found to be positively related to child overweight and obesity. Still, there is a gap in studies of factors influencing child overweight and obesity (over age 2 y) in the Pacific, especially those that use a longitudinal study design.

The risk factors consistently identified in this review align with those reported in other regions globally. The positive association between maternal BMI and childhood OWOB mirrors patterns documented in systematic reviews from high-income settings, where maternal adiposity has been consistently identified as a predictor of infant macrosomia and later obesity through intergenerational pathways. 79 The role of nutrition transition with a shift towards processed and energy-dense foods are consistent with findings from systemative reviews spanning Asian, Latin America, Europe, and the Pacific. 80 , 81 , 82 Western or modern dietary patterns were repeatedly associated with elevated obesity risk in children. 79 , 81 Screen time and childhood OWOB was consistent with global-wide literature linking sedentary media-based behavior to adiposity in children. 80 Similar patterns in risk factors such as screen time, poor diet quality, and parental obesity have been documented in the Middle East and North Africa, regions also undergoing rapid nutrition transition. 83 These parallels suggest that while the PICT has unique cultural and geographic features the modifiable drivers of childhood OWOB identified in this review are part of a broader global pattern associated with the nutrition transition.

While this systematic review uncovered several factors influencing childhood obesity in the Pacific islands, there are some limitations of this systematic review. Due to the differences in study settings and variation in how factors were measured, it is difficult to make direct comparisons between the studies. Some studies aggregated populations across Pacific Islands and other jurisdictions such as Hawai'i, Aotearoa (New Zealand), Australia. As these populations were outside the study's scope, their results were excluded, which may have further limited the findings of this review. Pacific Islands are relatively small populations, making sample size adequacy for statistical comparisons also challenging. Although an exhaustive literature search was conducted, there is possible selection bias

due to the timing of the search and the exclusion of gray literature. Additionally, publication bias may have limited the authors from finding studies that may have been implemented in the Pacific islands, but not published.

Applying the SEM to the findings highlights the influence of multilevel factors on overweight and obesity, and therefore the need for multi-level interventions to modify its occurrence. While individual behaviors like diet and physical activity are crucial, they are influenced by family dynamics, community structures, and socio-political forces which are vital, especially for long term impact. Further studies to understand factors at the policy level would also be beneficial as no studies were found to assess influences of policy on childhood OWOB in the Pacific islands.

Future interventions should integrate strategies targeting multiple SEM levels, such as improving school food environments, strengthening family-based nutrition education, and advocating for policies that promote affordable healthy food access.

In conclusion, despite being among the most affected by obesity, globally, the Pacific region remains understudied. Modern diet, screen time, maternal BMI, and child public school environment show evidence of positive association with child OWOB. There is an urgent need for multilevel, longitudinal research and interventions that address the immediate and upstream drivers of child OWOB. Nonetheless, limited evidence highlights the importance of both multilevel and life course approaches in the prevention of child overweight and obesity.

Contributors

Rachel Novotny conceived the study, served as a study screener and reviewer, oversaw screening, and wrote the final manuscript.

Rica Dela Cruz managed data for the study, served as primary study screener, reviewer and data compiler, wrote sections of the manuscript, and reviewed and approved the final manuscript.

Ashley B. Yamanaka contributed to study screening and reviewing, contributed to writing the manuscript, and reviewed and approved the final manuscript.

Fiorenzo Toncan screened studies and compiled data, and reviewed and approved the final manuscript.

Pauline McFall compiled data for the study, screened studies, and reviewed and approved the final manuscript.

Courtney C. Choy contributed to search methodology, screened and reviewed studies, contributed to revising the manuscript, and reviewed and approved the final manuscript.

Olivier Galy contributed to development of the study plan, and reviewed and approved the final manuscript.

Anthony Okely designed the child obesity series of studies, contributed to the development of the study plan, and reviewed and approved the final manuscript.

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Declaration of interests

RN is an associate editor for Nutrition Reviews; she received payment from ILSI for manuscript edit. AO is a member of the Research Advisory Committee for Health and Wellbeing Queensland, he has received consulting fees from World Bank and honorarium for a presentation at the Capital Institute of Pediatrics in Beijing. The rest of the authors declare no conflict of interest.